

Tracing Program Validasi HTML

Kelompok B6

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1.Pseudocode

Program : Main.c

```
// =====  
// MAIN PROGRAM (Validasi Tag HTML)  
// =====  
  
PROGRAM ValidasiTagHTML  
DEKLARASI  
s : String st :  
Stack  
hasTags : Boolean  
isValid : Boolean  
  
ALGORITMA  
PRINT "--- Validasi Tag HTML menggunakan Stack ---" PRINT  
"Masukkan string (ketik 'exit' untuk keluar):"  
  
WHILE TRUE DO INPUT s  
IF s == "exit" THEN BREAK END IF  
  
buatStack(&st) hasTags  
<- FALSE isValid <-  
TRUE i <- 0  
  
WHILE s[i] != '\0' DO IF s[i]  
== '<' THEN  
i <- i + 1 isClosing  
<- 0  
IF s[i] == '/' THEN isClosing <- 1 ; i <- i + 1 END IF  
  
Baca karakter hingga '>' ke tagName  
IF tagName kosong OR '>' tidak ditemukan THEN isValid <- FALSE BREAK END IF  
  
hasTags <- TRUE
```

```

IF NOT isClosing THEN
tambahStack(&st, tagName) ELSE
IF stackIsEmpty(&st) THEN isValid <- FALSE BREAK END IF topTag <-
hapusStack(&st)
IF topTag != tagName THEN isValid <- FALSE BREAK END IF END IF
END IF
i <- i + 1 END
WHILE

IF isValid AND hasTags AND stackIsEmpty(&st) THEN PRINT
"true"
ELSE
PRINT "false" END IF
END WHILE
END PROGRAM

```

Program : stack.c

```

// =====
// MODUL STACK (Validasi Tag HTML)
// =====

PROCEDURE buatStack(s : Pointer to Stack) s^.top
<- NULL
END PROCEDURE

FUNCTION stackIsEmpty(s : Pointer to Stack) -> Boolean RETURN
(s^.top == NULL)
END FUNCTION

PROCEDURE tambahStack(s : Pointer to Stack, nilai : String) ALOKASI
Memori untuk newNode
IF newNode == NULL THEN
PRINT "Error: Alokasi memori gagal." RETURN
END IF
newNode^.data <- nilai
newNode^.next <- s^.top s^.top
<- newNode
END PROCEDURE

FUNCTION hapusStack(s : Pointer to Stack) -> String IF
stackIsEmpty(s) THEN
RETURN "" END IF
buffer <- s^.top^.data temp <-

```

```

s^.top
s^.top <- s^.top^.next DEALOKASI
temp
RETURN buffer END
FUNCTION

PROCEDURE tampilkanStack(s : Pointer to Stack) IF
stackIsEmpty(s) THEN
PRINT " kosong " RETURN
END IF
curr <- s^.top
WHILE curr != NULL DO
PRINT curr^.data
IF curr^.next != NULL THEN PRINT " -> " END IF
curr <- curr^.next END
WHILE
PRINT (Baris Baru)
END PROCEDURE

```

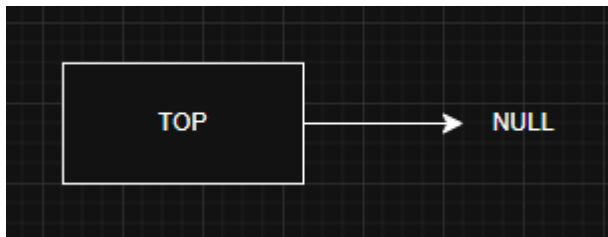
2.Tracing Program

No	Langkah	Operasi	Stack Sebelum	Stack Sesudah	Keterangan
1.	Inisialisasi	buatStack(&st)	[]	[]	
2.	Baca token <html>	tambahStack (&st, "html")	[]	[html]	
3.	Baca token <body>	tambahStack (&st, "body")	[html]	[body→html]	
4.	Baca token <h1>	tambahStack (&st, "h1")	[body → html]	[h1→body→html]	

5.	Baca token </h1>	topTag=hapusStack(&st) "h1"	[h1→body→html]	[body→html]	
6.	Baca token </body>	topTag=hapusStack(&st) "body"	[body→html]	[html]	
7.	Baca token </html>	topTag=hapusStack(&st) "html"	[html]	[]	
8.	Evaluasi	isValid=true hasTags=true stackIsEmpty=true	[]	[]	

3. Visualisasi

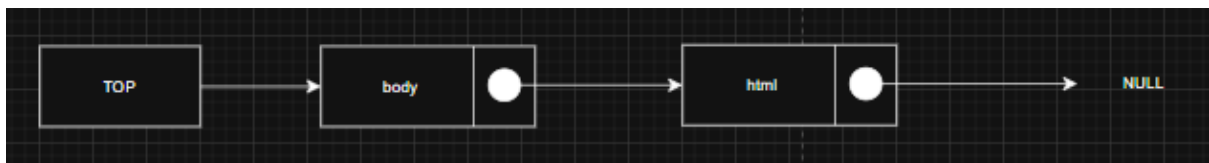
Langkah 1: Inisialisasi (buatStack)



Langkah 2: Tambah <html> push "html"



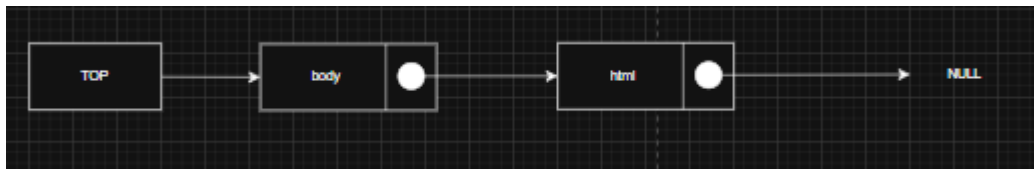
Langkah 3 : Tambah <body> push "body"



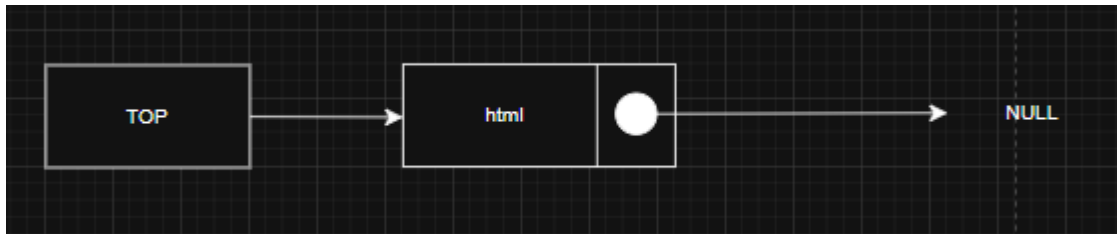
Langkah 4: Tambah <h1> push "h1"



Langkah 5: Tutup </h1> pop "h1" (cocok ✓)



Langkah 6: Tutup `</body>` pop "body"



Langkah 7: Tutup `</html>` pop "html"

